The waist class is quadratic, not additive, under band composition on T^2

MacBeth
Kodamai

2026-10-14

Author: MacBeth (Kodamai). **Date:** 2026-10-14.
Prepared for Rick (referee) and Neil Ghani (PI); **CC** Robin Langer.
Content commit: `16b051d` (supersedes `3954636`).
Provenance. Registry node `delta-additive-in-band` [**proved**] in registry `supply-chain-inventory-gerbe-h2`; child of `coupled-T2-general-delta-lemma` [**proved**] and `gerbe-two-ends-correlation` [**proved**]. Proof file `proofs/2026-10-13-delta-additive-in-band.md`; verification script `scratch/2026-10-13-delta-additivity.py` (couplers $D_4 \times C_2$, $Q_8 \times C_2$; 0 violations on every clause below). This note typesets a complete, registry-graded result. Its one import is Theorem 7 (the common receptacle off Out^0) together with the subgroup counts 0/10304, 0/36420 and the "14 of ..." / "50 of ..." enumerations: these are R. (Rick)'s computed and by-hand results of 2026-10-14, reproduced here with credit. All other claims are from the cited proof file.

Abstract

Fix base $C = B(\mathbb{Z}^2)$ (nerve T^2) and a structure group K of nilpotency class ≤ 2, so that $M := \mathrm{Inn}K$ is abelian. The general δ-commutator lemma assigns to a band $\rho\colon \mathbb{Z}^2 \to \mathrm{Out}K$ — a commuting pair of outer classes — the connecting class $\delta(\rho) = [[A, B]] \in H^2(T^2; M)$ for any lifts $A, B \in \mathrm{Aut}K$. We determine how δ behaves under band composition. The answer is: *not additively.* A master cross-term formula (Theorem 1), an on-the-nose identity in $\mathrm{Inn}K$, computes the composite commutator in full. On the trivial-action locus Out^0, where the additivity question is well-typed, this collapses to a quadratic law $\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \chi$ with symmetric bi-additive polar form χ (Theorem 5), whence $\delta(\rho^n) = n^2\delta(\rho)$. A dichotomy (Theorem 6) shows that *for an abelian subgroup $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$ δ is additive on $\bar{\mathcal{A}}$ iff it is identically zero there: on the trivial-action locus there is no abelian subgroup on which δ is a nonzero homomorphism*, so a coupled torus in Out^0 is always a witness to non-additivity. (The scope $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$ is essential: off Out^0 this can fail — there are abelian subgroups, 1 for $D_4 \times C_2$ and 3 for $Q_8 \times C_2$, each of order 8 meeting Out^0 in order 4, that contain a coupled torus $(a, b) \subseteq \mathrm{Out}^0$ with $\delta(a, b) \neq 0$ yet on which the induced $\bar{\delta}$ is additive.) On Out^0, coupling, non-splitting of the lift extension, and non-additivity of the waist class are one and the same phenomenon; this is the classical commutator form of the central extension Aut^0K — the $\mathrm{Hom}(\Lambda^2\bar{\mathcal{A}}, M)$ component of H^2. All clauses are verified computationally with 0 violations on the two T^2 couplers.

1 The question and the answer

Fix base $C = B(\mathbb{Z}^2)$ (nerve T^2) and a structure group K of nilpotency class ≤ 2, so $M := \mathrm{Inn}K$ is abelian. By the general δ-commutator lemma (`coupled-T2-general-delta-lemma`, proved 2026-10-12), a band $\rho\colon \mathbb{Z}^2 \to \mathrm{Out}K$ — i.e. a commuting pair $\bar{A} := \rho(s)$, $\bar{B} := \rho(t) \in \mathrm{Out}K$ — has

connecting class

$$\delta(\rho) = [\,[A,B]\,] \in H^2(T^2;M) = M_{\mathbb{Z}^2} = M\big/\big((\sigma-1)M + (\tau-1)M\big), \tag{1}$$

where $A, B \in \mathrm{Aut}K$ are *any* lifts of $\bar{A}, \bar{B}$ and $\sigma = \bar{A}\cdot(-)$, $\tau = \bar{B}\cdot(-)$ are the induced actions of $\bar{A}, \bar{B}$ on M by conjugation.

Let ρ_1, ρ_2 be two bands landing in a *common abelian subgroup* $\bar{\mathcal{A}} \leq \mathrm{Out}K$ (the concrete meaning of "a common system of commuting lifts": their pointwise product $\rho_1\rho_2 = (\bar{A}_1\bar{A}_2,\ \bar{B}_1\bar{B}_2)$ is again a band into $\bar{\mathcal{A}}$, because $\bar{\mathcal{A}}$ is abelian). **Is $\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2)$?**

Answer: (B) — No. δ is not a homomorphism. The exact defect is governed by the master cross-term formula (Theorem 1), an on-the-nose identity in $\mathrm{Inn}K$:

$$[A_1A_2,\ B_1B_2] \;=\; \underbrace{[A_1,B_1]}_{=\,\delta(\rho_1)\ \text{rep}} \;+\; (\sigma_1\tau_1)\cdot[A_2,B_2] \;+\; \sigma_1\cdot[A_2,B_1] \;+\; \tau_1\cdot[A_1,B_2].$$

On the *trivial-action locus* $\mathrm{Out}^0 = \ker(\mathrm{Out}K \to \mathrm{Aut}M)$ — where the question is even well-typed, all three classes living in the *same* group M — this collapses (Theorem 5) to

$$\boxed{\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \chi(\rho_1,\rho_2), \qquad \chi(\rho_1,\rho_2) = [A_1,B_2] + [A_2,B_1] \in M}$$

with χ a *symmetric bi-additive* form — the polarization of the quadratic map δ, which obeys $\delta(\rho^n) = n^2\,\delta(\rho)$. And then the *dichotomy* (Theorem 6): on any abelian $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$,

$$\delta \text{ is additive on bands into } \bar{\mathcal{A}} \iff \delta \equiv 0 \text{ there} \iff \pi^{-1}(\bar{\mathcal{A}}) \leq \mathrm{Aut}K \text{ is abelian.}$$

On the trivial-action locus Out^0 there is no abelian subgroup on which δ is a nonzero homomorphism. A coupled torus in Out^0 is *always* a witness to non-additivity. (This is sharp to the locus: off Out^0 there exist abelian $\bar{\mathcal{A}}$ — 1 for $D_4 \times C_2$, 3 for $Q_8 \times C_2$, each of order 8 with $|\bar{\mathcal{A}} \cap \mathrm{Out}^0| = 4$ — that contain a coupled torus $(a,b) \subseteq \mathrm{Out}^0$, $\delta(a,b) \neq 0$, yet on which the induced $\bar{\delta}$ is additive, the elements outside Out^0 killing $\delta(a,b)$ in coinvariants.)

So the butterfly §7 open edge "composite weld obstruction = composite of waist classes?" resolves in the negative, in the sharpest possible form: the waist class δ is a *quadratic refinement*, not a homomorphism, on the composition monoid of bands; its polar form χ is the obstruction to additivity, with $\chi|_{\bar{\mathcal{A}}} \equiv 0$ iff $\delta|_{\bar{\mathcal{A}}} \equiv 0$ for every abelian $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$.

2 The master cross-term formula

Throughout, $M = \mathrm{Inn}K \trianglelefteq \mathrm{Aut}K$ is abelian (class ≤ 2), written additively, and for $X \in \mathrm{Aut}K$, $m \in M$ we write $X \cdot m := XmX^{-1} \in M$ (M normal); this factors through $\pi(X) \in \mathrm{Out}K$. For a band ρ_i fix lifts $A_i, B_i \in \mathrm{Aut}K$ of $\rho_i(s), \rho_i(t)$ and set $\sigma_i := \rho_i(s)\cdot(-)$, $\tau_i := \rho_i(t)\cdot(-) \in \mathrm{Aut}(M)$.

Theorem 1 (master cross-term formula)**.** *Let K be finite of class ≤ 2 and let ρ_1, ρ_2 be bands with all four outer classes $\bar{A}_1, \bar{B}_1, \bar{A}_2, \bar{B}_2$ in a common abelian subgroup of $\mathrm{Out}K$. Then every pairwise commutator $[A_i, B_j]$ lies in M, and, additively in M,*

$$[A_1A_2,\ B_1B_2] = [A_1,B_1] + (\sigma_1\tau_1)\cdot[A_2,B_2] + \sigma_1\cdot[A_2,B_1] + \tau_1\cdot[A_1,B_2]. \tag{$\star$}$$

Since A_1A_2, B_1B_2 are lifts of $(\rho_1\rho_2)(s), (\rho_1\rho_2)(t)$, the left side represents $\delta(\rho_1\rho_2) \in M_{\mathbb{Z}^2}$ (coinvariants for the product action $\sigma_1\sigma_2, \tau_1\tau_2$).

Proof. Membership: $\pi[A_i, B_j] = [\bar{A}_i, \bar{B}_j] = 1$ because $\bar{A}_i, \bar{B}_j$ commute (common abelian subgroup), so $[A_i, B_j] \in \ker \pi = M$. Use the universal commutator identities

$$[xy, z] = x[y, z]x^{-1} \cdot [x, z], \qquad [x, yz] = [x, y] \cdot y[x, z]y^{-1}.$$

Apply the first with $x = A_1, y = A_2, z = B_1B_2$:

$$[A_1A_2, B_1B_2] = A_1[A_2, B_1B_2]A_1^{-1} \cdot [A_1, B_1B_2].$$

Apply the second to each factor:

$$[A_2, B_1B_2] = [A_2, B_1] \cdot B_1[A_2, B_2]B_1^{-1}, \qquad [A_1, B_1B_2] = [A_1, B_1] \cdot B_1[A_1, B_2]B_1^{-1}.$$

Substituting and conjugating the first bracket by A_1:

$$[A_1A_2, B_1B_2] = \big(A_1[A_2, B_1]A_1^{-1}\big) \cdot \big(A_1B_1[A_2, B_2]B_1^{-1}A_1^{-1}\big) \cdot [A_1, B_1] \cdot \big(B_1[A_1, B_2]B_1^{-1}\big).$$

All four factors lie in the abelian group M, so we may reorder and pass to additive notation. The conjugating prefixes are A_1 (giving σ_1), A_1B_1 (giving $\sigma_1\tau_1$), nothing (giving 1), and B_1 (giving τ_1). This is exactly $(\star)$. □

Verification. Script `scratch/2026-10-13-delta-additivity.py`, clause `[GEN]`: $(\star)$ holds *on the nose in* $\mathrm{Inn}K$ over *all* quadruples of bands inside a common abelian subgroup — 8113 checks for $D_4 \times C_2$ and 26037 for $Q_8 \times C_2$, 0 **violations**. (An earlier run comparing against the representative lift of the *product coset* showed spurious "violations"; those are precisely coboundaries $(\sigma - 1)M + (\tau - 1)M$ — the product-coset representative differs from A_1A_2 by an inner automorphism — and vanish in $M_{\mathbb{Z}^2}$. Using the on-the-nose product lift A_1A_2 removes them, confirming $(\star)$ is an identity in M, not merely in the quotient.)

Reading of $(\star)$. Beyond the two "diagonal" terms $\delta(\rho_1), \delta(\rho_2)$ there are: a *twist* $(\sigma_1\tau_1)$ acting on $\delta(\rho_2)$'s representative, and the two *cross terms* $\sigma_1[A_2, B_1] + \tau_1[A_1, B_2]$. Both the twist and the asymmetry (σ_1, τ_1 but not σ_2, τ_2) are artifacts of writing the product lift as A_1A_2 rather than A_2A_1; they disappear precisely when the action is trivial.

3 The trivial-action locus: δ is a quadratic map

Definition 2. $\mathrm{Out}^0 := \ker\big(\mathrm{Out}K \to \mathrm{Aut}(M)\big)$ is the subgroup of outer classes acting trivially on $M = \mathrm{Inn}K$. A band ρ is *trivial-action* iff $\bar{A}, \bar{B} \in \mathrm{Out}^0$, i.e. $\sigma = \tau = \mathrm{id}$.

Remark 3 (terminology)**.** The PROVE brief's "trivial centre-action" is here made precise as the action relevant to δ: trivial action of $\mathrm{Out}K$ on the coefficient module $M = \mathrm{Inn}K$ — the exact condition under which δ's receptacle $M_{\mathbb{Z}^2}$ collapses to M and the additivity question becomes well-typed. For class-≤ 2 K this is also the condition making the restricted extension $(*)|_{\mathrm{Out}^0}$ central, below. We do *not* silently conflate it with the distinct condition "$\mathrm{Out}K$ acts trivially on $Z(K)$"; the relevant module for the waist class is $\mathrm{Inn}K$, not $Z(K)$.

Over Out^0 the restricted extension $1 \to M \to \mathrm{Aut}^0K \to \mathrm{Out}^0 \to 1$ (with $\mathrm{Aut}^0K := \pi^{-1}(\mathrm{Out}^0)$) is a *central* extension: M is central in Aut^0K (verified: clause `restricted (*) central: True` for both couplers). For such a central extension the *commutator pairing*

$$c\colon \{\text{commuting pairs in } \mathrm{Out}^0\} \to M, \qquad c(\bar{x}, \bar{y}) := [\tilde{x}, \tilde{y}]$$

is well defined (lift-independent: for central $u, v \in M$, $[\tilde{x}u, \tilde{y}v] = [\tilde{x}, \tilde{y}]$) and $\delta(\rho) = c(\rho(s), \rho(t))$ for trivial-action bands, with $M_{\mathbb{Z}^2} = M$ (no coboundaries).

Lemma 4 (c is alternating and bi-multiplicative)**.** *On commuting pairs in* Out^0*:* $c(\bar{x},\bar{x}) = 0$*;* $c(\bar{y},\bar{x}) = -c(\bar{x},\bar{y})$*; and whenever all pairs in sight commute,*

$$c(\bar{x}\bar{y},\bar{z}) = c(\bar{x},\bar{z}) + c(\bar{y},\bar{z}), \qquad c(\bar{x},\bar{y}\bar{z}) = c(\bar{x},\bar{y}) + c(\bar{x},\bar{z}).$$

Proof. Alternating/antisymmetry are immediate from $[\tilde{x},\tilde{x}] = 1$, $[\tilde{y},\tilde{x}] = [\tilde{x},\tilde{y}]^{-1}$. For bi-multiplicativity,

$$[\tilde{x}\tilde{y},\tilde{z}] = \tilde{x}[\tilde{y},\tilde{z}]\tilde{x}^{-1}\,[\tilde{x},\tilde{z}] = [\tilde{y},\tilde{z}][\tilde{x},\tilde{z}],$$

using that $[\tilde{y},\tilde{z}] \in M$ is *central* (this is exactly the trivial-action hypothesis). □

(Clause `(2) bi-mult on Out^0`: 0 violations; and `(2)` over *all* of Out **fails** — 112/1408 and 962/4032 violations — confirming centrality is essential.)

Theorem 5 (quadratic law + cross term)**.** *For trivial-action bands* ρ_1, ρ_2 *into a common abelian* $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$*, equality holds in* M*:*

$$\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \chi(\rho_1,\rho_2), \qquad \chi(\rho_1,\rho_2) = c(\bar{A}_1,\bar{B}_2) + c(\bar{A}_2,\bar{B}_1) = [A_1,B_2] + [A_2,B_1].$$

χ *is symmetric (*$\chi(\rho_1,\rho_2) = \chi(\rho_2,\rho_1)$*) and bi-additive;* δ *is a* quadratic map $\mathrm{Hom}(\mathbb{Z}^2,\bar{\mathcal{A}}) \cong \bar{\mathcal{A}}\times\bar{\mathcal{A}} \to M$ *with polar form* χ*, and in particular* $\delta(\rho^n) = n^2\,\delta(\rho)$ *and* $\delta(\rho^{-1}) = \delta(\rho)$*.*

Proof. Either specialize ($\star$) to $\sigma_1 = \tau_1 = \mathrm{id}$, or compute directly with Lemma 4:

$$\delta(\rho_1\rho_2) = c(\bar{A}_1\bar{A}_2,\bar{B}_1\bar{B}_2) = c(\bar{A}_1,\bar{B}_1) + c(\bar{A}_1,\bar{B}_2) + c(\bar{A}_2,\bar{B}_1) + c(\bar{A}_2,\bar{B}_2) = \delta(\rho_1) + \delta(\rho_2) + \chi.$$

Symmetry of χ is visible; bi-additivity is bi-multiplicativity of c. Quadraticity: $\delta(\rho^n) = c(\bar{A}^n,\bar{B}^n) = n^2c(\bar{A},\bar{B}) = n^2\delta(\rho)$ by bi-multiplicativity; $\delta(\rho^{-1}) = c(\bar{A}^{-1},\bar{B}^{-1}) = (-1)(-1)c(\bar{A},\bar{B}) = \delta(\rho)$. □

Verification. Clause `(3) cross-term on Out^0` (1905 checks each) and `(4) quadratic` (320 checks each, $n \le 6$): 0 **violations** on both couplers.

4 The additivity dichotomy

Theorem 6 (dichotomy)**.** *Let* $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$ *be abelian. The following are equivalent:*

1. δ *is a group homomorphism on* $\mathrm{Hom}(\mathbb{Z}^2,\bar{\mathcal{A}})$ *(additivity);*
2. $\delta \equiv 0$ *on all bands into* $\bar{\mathcal{A}}$*;*
3. $c|_{\bar{\mathcal{A}}\times\bar{\mathcal{A}}} \equiv 0$*;*
4. $\pi^{-1}(\bar{\mathcal{A}}) \le \mathrm{Aut}K$ *is abelian (i.e.* $\bar{\mathcal{A}}$ *has a system of commuting lifts).*

In particular δ *is* never a nonzero homomorphism *on any abelian subgroup.*

Proof. (2) ⇒ (1) trivial. (1) ⇒ (3): additivity forces $\chi \equiv 0$; take $\rho_1 = (\bar{x},1), \rho_2 = (1,\bar{y})$, so $\delta(\rho_1) = c(\bar{x},1) = 0$, $\delta(\rho_2) = c(1,\bar{y}) = 0$, while $\rho_1\rho_2 = (\bar{x},\bar{y})$ gives $\delta(\rho_1\rho_2) = c(\bar{x},\bar{y})$; additivity makes this 0 for all $\bar{x},\bar{y} \in \bar{\mathcal{A}}$. (3) ⇒ (2): $\delta(\rho) = c(\bar{A},\bar{B}) = 0$. (3) ⇔ (4): $\pi^{-1}(\bar{\mathcal{A}})$ is a central extension of the abelian $\bar{\mathcal{A}}$ by M, hence abelian iff its commutator pairing $c|_{\bar{\mathcal{A}}}$ vanishes. □

Verification. Clause `(5) dichotomy on Out^0`: among the 6 coupled abelian subgroups of Out^0 (for each coupler), the count `nonzero & ADDITIVE` is 0; every coupled subgroup is non-additive. Exhaustive over ≤ 2-generated abelian subgroups (which suffices, bands being rank ≤ 2).

Reconciliation with the coupling criterion. (4) is the negation of the δ-lemma's coupling criterion "$\exists$ commuting outer pair with no commuting lift." So Theorem 6 says, *for abelian* $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$:

$$\bar{\mathcal{A}} \text{ couples (some band has } \delta \neq 0) \iff \pi^{-1}(\bar{\mathcal{A}}) \text{ nonabelian} \iff \delta \text{ is non-additive on } \bar{\mathcal{A}}.$$

On the trivial-action locus, coupling, non-splitting of $\pi^{-1}(\bar{\mathcal{A}})$, and non-additivity of the waist class are *one and the same phenomenon* — this is the classical commutator form of a central extension, the $\mathrm{Hom}(\Lambda^2\bar{\mathcal{A}}, M)$ component of $H^2(\bar{\mathcal{A}}; M)$, and we claim no novelty for it here. Off Out^0 the chain breaks at (4): $\pi^{-1}(\bar{\mathcal{A}})$ need not be central over $\bar{\mathcal{A}}$, and (Theorem 7) an abelian $\bar{\mathcal{A}} \not\subseteq \mathrm{Out}^0$ can contain a coupled torus while $\bar{\delta}$ is additive on it (1 such $\bar{\mathcal{A}}$ for $D_4 \times C_2$, 3 for $Q_8 \times C_2$).

5 Scope, and the genuinely twisted regime

The clean additivity package (Theorems 5–6) lives on the *trivial-action locus* Out^0, and this is not a cosmetic restriction but the precise place the question is *well-typed*: for bands with differing actions, $\delta(\rho_1), \delta(\rho_2), \delta(\rho_1\rho_2)$ lie in *different* coinvariant quotients $M_{(\sigma,\tau)}$, so "$\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2)$" is not even a statement about one group until a common receptacle is chosen.

Coupling is *not* confined to Out^0. The census (`coupling vs action`) finds coupled bands ($\delta \neq 0$ in the quotient) with *nontrivial* action: 24 of 60 for $D_4 \times C_2$, 72 of 108 for $Q_8 \times C_2$. For these, the governing statement is the master formula ($\star$) alone — a correct on-the-nose identity in $\mathrm{Inn}K$. Its descent to a common coinvariant quotient does *not*, however, retain the twist: in the full coinvariant module the factor $(\sigma_1\tau_1 - 1)\cdot\delta(\rho_2)$ lies in the augmentation submodule $\sum_x (x-1)M$, so the twist *dies* in that quotient (it is computed nonzero in 0/10304 and 0/36420 checks). The honest common receptacle is therefore the subgroup-coinvariant module itself:

$$M_{\bar{\mathcal{A}}} := M \Big/ \textstyle\sum_{x \in \bar{\mathcal{A}}} (x-1)M,$$

the coinvariants for the whole abelian subgroup $\bar{\mathcal{A}} = \langle \bar{A}_1, \bar{B}_1, \bar{A}_2, \bar{B}_2 \rangle$ generated by the four outer classes.

Theorem 7 (common receptacle off Out^0)**.** *Let $\bar{\mathcal{A}} \leq \mathrm{Out}K$ be abelian (not necessarily $\subseteq \mathrm{Out}^0$). In $M_{\bar{\mathcal{A}}}$ the induced commutator pairing $\bar{c}$ is well defined and bi-additive, and Theorem 5 together with the equivalences* (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3) *of Theorem 6 hold* verbatim *with $\bar{c}, \bar{\delta}$ in place of c, δ. The remaining equivalence "*(1)–(3) $\Leftrightarrow$ (4)*" [$\pi^{-1}(\bar{\mathcal{A}})$ abelian] is the one clause that* fails *off* Out^0.

Verified: 0 violations on *all* 14 abelian subgroups of $\mathrm{Out}(D_4 \times C_2)$ and all 50 of $\mathrm{Out}(Q_8 \times C_2)$ that are not $\subseteq \mathrm{Out}^0$. (Theorem 7 and these enumerations are R.'s computed and by-hand results of 2026-10-14.) This closes — rather than defers — the common-receptacle edge: the only genuine loss off Out^0 is clause (4), since the lift extension $\pi^{-1}(\bar{\mathcal{A}})$ need no longer be a central extension of $\bar{\mathcal{A}}$.

- **No gap in the posed question.** The question fixes trivial centre-action; there the answer is complete and proved: (B), with $\chi = [A_1, B_2] + [A_2, B_1]$, quadratic law, dichotomy.

6 Verification summary (hostile-referee pass)

All on `scratch/2026-10-13-delta-additivity.py`, couplers $D_4 \times C_2$ ($|\mathrm{Out}| = 16$), $Q_8 \times C_2$ ($|\mathrm{Out}| = 48$):

clause	statement	checks ($D_4\times C_2$ / $Q_8\times C_2$)	violations
`[GEN]`	master formula ($\star$) on-the-nose in M	8113 / 26037	**0 / 0**
central	($*$) restricted to Out^0 is central	—	True / True
`(2)`	c bi-multiplicative on Out^0	512 / 512	**0 / 0**
`(2)` over all Out	bi-multiplicativity **fails** off Out^0	1408 / 4032	112 / 962 *(expected)*
`(3)`	$\delta(\rho_1\rho_2) = \delta_1 + \delta_2 + \chi$ on Out^0	1905 / 1905	**0 / 0**
`(4)`	$\delta(\rho^n) = n^2\delta(\rho)$ on Out^0	320 / 320	**0 / 0**
`(5)`	dichotomy: nonzero & additive	6 coupled subs each	**0 / 0**

- **Theorem 1 — GREEN** (hand proof from universal commutator identities + M abelian; computed confirmation 0 violations). The only hypotheses are class ≤ 2 and common abelian subgroup.
- **Theorems 5–6 — GREEN** (two-line proofs from the central commutator pairing; computed confirmation).
- **Boundary cases.** Trivial band $\rho = 1$: $\chi(1,\rho) = c(1,\bar{B}) + c(\bar{A},1) = 0$, additive ✓. $\rho_1 = \rho_2$: $\chi(\rho,\rho) = 2c(\bar{A},\bar{B}) = 2\delta(\rho)$, consistent with $\delta(\rho^2) = 4\delta(\rho)$ ✓.
- **No RED. No YELLOW** inside the stated scope; the twisted edge of §5 is flagged and is not part of the posed question.

7 Grant / program placement

This is the *composition-monoid structure of the waist class*. The butterfly / weak-morphism arc asked whether the weld obstruction composes additively; the answer sharpens the whole gerbe lane:

- **Theory.** The waist class δ is a *quadratic refinement* on the band composition monoid, with explicit polar form χ = the commutator pairing of the central extension $\mathrm{Aut}^0 K$. This is the T^2 counterpart of the floor D_1/D_2 dichotomy and of the two-ends forced correlation: on Out^0, "coupling = nonabelian lift = non-additivity" is a single statement.
- **Applications.** For orchestration / ZS welds, composing two welds does *not* add their obstruction classes; the correction χ is an economically real interaction term (the "cross-coupling" of two independently-welded subsystems). Additivity of the composite, $\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2)$, holds iff the cross-term $\chi(\rho_1,\rho_2) = c(\bar{A}_1,\bar{B}_2) + c(\bar{A}_2,\bar{B}_1)$ vanishes on the *pair* — *not* iff each subsystem is individually uncoupled. Indeed $\rho_1 = (\bar{x},1)$, $\rho_2 = (1,\bar{y})$ have $\delta(\rho_1) = \delta(\rho_2) = 0$ yet $\delta(\rho_1\rho_2) = c(\bar{x},\bar{y})$, nonzero whenever $\bar{x},\bar{y}$ couple: two individually-uncoupled welds can interact.
- **Formalisation.** Theorem 1's identity ($\star$) and Lemma 4 are finite, decidable, and a natural Lean target over the existing Inn/Out engine.

One-line summary. On T^2 the band-to-waist connecting map is a *quadratic map*, not a homomorphism: in general $[A_1A_2, B_1B_2] = [A_1, B_1] + (\sigma_1\tau_1)[A_2, B_2] + \sigma_1[A_2, B_1] + \tau_1[A_1, B_2]$, which on the trivial-action locus reads $\delta(\rho_1\rho_2) = \delta(\rho_1) + \delta(\rho_2) + \big([A_1, B_2] + [A_2, B_1]\big)$ with symmetric bi-additive polar form χ; and *for an abelian subgroup* $\bar{\mathcal{A}} \subseteq \mathrm{Out}^0$ δ is additive on $\bar{\mathcal{A}}$ **iff** it is identically zero there — a coupled torus in Out^0 is always a witness to non-additivity. (Off Out^0 the induced $\bar{\delta}$ can be additive on an abelian subgroup that still contains a coupled torus: 1 such subgroup for $D_4 \times C_2$, 3 for $Q_8 \times C_2$.)